



PPC-6001
MAY 6, 2014



PPC6000 SERIES FUEL AIR RATIO CONTROLLER

DESCRIPTION

The Fireye PPC6000 is a state of the art stand-alone parallel positioning system for all types of liquid or gaseous fuel fired combustion systems. When combined with a Fireye flame safeguard system such as the Fireye BurnerLogix control, the PPC6000 offers the most compact and advanced parallel positioning system available. Four fuel profiles allow the PPC6000 to accommodate a variety of applications such as two fuels, with and without, the optional variable speed drive (VSD). With each profile having up to 24 points entered to assure a smooth "curve", the microprocessor interpolates points between entered values and precisely positions fuel and air to within $\pm .1$ degree. This eliminates hysteresis for improved efficiency. The PPC6000 is capable of powering up to ten 24volt servomotors supplied from the integral transformer. All servomotors and displays operate on a secure CANbus communications protocol and can be "daisy chained" together for simplified wiring. Two independent PID control loops for temperature or pressure control provide precise, accurate control of firing rate for unmatched response to load changes. Four safety rated user definable digital inputs are standard and can be configured for functions such as lead lag, night setback, thermal shock warm up, etc. Digital inputs can also be configured as analog inputs via the optional programmable function blocks using Fireye Abacus software. Built in lead lag sequencing for up to four boilers is included in every PPC6000. More elaborate lead lag schemes may be possible using the programmable function blocks.

Two variations of multifunction displays are available:

- NXTSD104 10.4" Color Touchscreen display with Ethernet, Modbus TCP/IP, 4 programmable relays and backup.
- NX610 display provides automatic storage of operating parameters and profiles for upload/download to the PPC6000. In addition, three line voltage fully programmable relays are mounted inside the display.



Other options include, two Variable speed drive (VSD) outputs via daughter board, Oxygen Trim using Fireye's proven heated zirconia oxide probe and optional interface, a wide variety of communications platforms such as Modbus is also available.

An optional internal freely programmable logic section of the PPC6000 means that external PLC's and expensive stand-alone sequencing controls may no longer be needed. Easy to program and protect, this new feature will make the PPC6000 suitable for many unique or unusual applications. Functions such as lead lag, sequencing, feedwater control and draft control are just some of the ways this feature can be used.

Optional ComFire software allows the user to view and control all aspects of the burner system, provides real time trending and logging, upload and download of profiles and options. The user can build custom screens or import actual photographs of the combustion equipment for animation.

Summary of PPC6000 Features

- State of the art surface mount technology
- Smallest footprint available: 7.0" x 4.5" x 2.5"
- Capable up to ten servomotors powered from the control
- Two PID loops for precise process control
- Simple four wire CANbus wiring reduces time and mistakes
- Optional VSD card provides two VSD's and one Analog output
- Re-transmittal of most data as 420mA output (requires NXDBVSD option)
- VSD encoder feedback available
- Standby lag boiler on water temp.
- Built in lead/lag sequencing for up to four boilers
- Fully programmable function block programming allows for custom applications such as draft control or feedwater control
- Two levels of displays: 12 Key Keypad with Backup, 10.4" Color Touchscreen
- Unparalleled worldwide support

Safety information



WARNING

The equipment described in this manual is capable of causing property damage, severe injury, or death. It is the responsibility of the owner or user to ensure that the equipment described herein is installed, operated and commissioned in compliance with the requirements of all national and local legislation, which may prevail.

When this equipment is fitted to an appliance due regard must also be given to the requirements of that appliance.

Before attempting to install, commission or operate this equipment all relevant sections of this document must be read and fully understood. If in doubt about any requirements consult Fireeye.

Installation, commissioning or adjustment of this product MUST be carried out by suitably trained engineers or personnel qualified by training and experience.

After installation or modifications to the installation all functions of the equipment MUST be checked to ensure safe and reliable operation of the control.

The manufacturer of this equipment accepts no liability for any consequences resulting from inappropriate, negligent or incorrect installation, commissioning or adjustment of operating parameters of the equipment. Control panels **must not** be left uncovered while power is on. If it is essential to do so while rectifying faults only personnel qualified by training and experience should be involved.

The time any covers are off must be kept to a minimum and warning notices **must** be posted.

Before attempting any work on this equipment or any equipment controlled by or connected to this equipment, all related electrical supplies **must** be isolated.

Safety interlocks **must not** be removed or over-ridden. Any faults once detected **must** be corrected before the control is operated.

**CAUTION – SOME VERSIONS OF THIS EQUIPMENT CONTAIN
A LITHIUM BATTERY IN THE DISPLAY UNIT**

NOTE

The manufacturer of this equipment has a policy of continual product improvement and reserves the right to change the specification of the equipment and the contents of this manual without notice.



Table of contents

This manual describes the installation, commissioning, operation and maintenance of the PPC6000 series fuel air ratio controls. It may be used in conjunction with the following other manuals:

- NEX-1502 - ComFire combustion analysis tool user manual

Please read the safety information at the front of this manual before proceeding.

1	Technical specification	11
1.1	General	11
1.2	Servo-motor control – IMPORTANT: See Section 3.2	11
1.3	Digital outputs (<i>PPC6000</i>)	12
1.4	Alarm output (<i>PPC6000</i>)	12
1.5	Digital inputs (<i>PPC6000</i>)	13
1.6	Pressure/Temperature Input (<i>PPC6000</i>)	13
1.7	Communications interface (<i>PPC6000</i>)	13
1.8	Optional Oxygen Probe Interface Unit (<i>NXO2INT</i>) - optional	14
1.9	Optional Ambient Air Temperature Unit (<i>NXIATS</i>) - optional	14
1.10	Variable Speed Drive (VSD) Daughter Board (<i>NXDBVSD</i>) - optional	14
1.11	Fireeye NXC04, NXC12, NXC20, NXC40 Servo Motors	15
1.12	Fireeye NXO2INT Oxygen Probe Interface Unit (optional)	15
1.13	Fireeye NXIATS Ambient Air Temperature Sensor (optional)	16
1.14	Approvals	16
1.15	Parts List with Description	16
2	Installation	18
2.1	Mounting details for the PPC6000 control	19
2.2	Mounting details for the display module	20
2.3	Mounting details for optional Oxygen Probe Interface unit, NXO2INT	21
2.4	The oxygen trim option	22
2.4.1	Oxygen Probe description	22
2.4.2	Installation of oxygen probe	22
2.4.3	Mounting the oxygen probe	23
2.4.4	Mounting arrangements for Temperature and Steam pressure sensors	24
2.4.5	Boiler temperature sensors	24
2.4.6	Steam pressure sensors	25
2.5	Option link selection (<i>PPC6000</i>)	26
2.5.1	General (Access to jumper's and fuse's is gained by removing the back cover of the PPC6000)	26
2.5.2	Line supply voltage (LK1 - 4) (<i>PPC6000</i>)	26
2.5.3	SENS IN and SENS SUPP (boiler temp/pressure sensor) (<i>PPC6000</i>)	27
2.5.4	REMOTE SETPOINT (<i>PPC6000</i>)	27
2.5.5	RS485 serial communications termination resistor (<i>PPC6000</i>)	27
2.5.6	RS485 serial communications termination resistor (<i>daughter board</i>)	27
	IN	27
	IN	27
	IN	27
	IN	27
2.6	Wiring	29
2.6.1	General	29

2.6.2	Grounding cable screens	32
2.6.3	GROUND (EARTH) connection	32
2.6.4	Ground (earth) Connection (<i>display unit</i>)	33
2.6.5	Terminal Designation	33
2.6.6	LINE and NEUTRAL supply (<i>PPC6000</i>) PE1 & PE2	35
2.6.7	Safety Shutdown output (<i>PPC6000</i>)	36
2.6.8	Controlled Shutdown Relay Output (<i>PPC6000</i>)	36
2.6.9	Alarm Relay output – RELAY 4 (<i>on the PPC6000</i>)	36
2.6.10	Auxiliary Relay Outputs (<i>display</i>)	37
2.6.11	Fuel Select /Profile Select Input/Purge and Ignition Prove Outputs (<i>PPC6000</i>)	37
2.6.12	Low Voltage Digital Inputs (<i>PPC6000</i>) – relevant Engineer's Keys EK1-EK4 (Sec. 6.6.2)	38
2.6.13	Servo-motor and Display Connection (<i>PPC6000</i>)	38
2.6.14	Display Connection with PPC6000 and BurnerLogix YB110	39
2.6.15	Oxygen Probe Interface Connection (<i>PPC6000</i>)	40
2.6.16	Pressure/temperature sensor input (<i>PPC6000</i>)	40
2.6.17	Remote Setpoint (Analog Input 5) (<i>PPC6000</i>)	41
2.6.18	RS485 interface (<i>PPC6000</i>)	41
2.6.19	Variable Speed Drive Selection:	42
2.6.20	PPC6000 Variable Speed Drive Daughter Board P/N NXDBVSD	43
2.6.21	LIVE and NEUTRAL supply (<i>optional Oxygen Probe Interface unit</i>)	44
2.6.22	Oxygen probe connection (<i>optional Oxygen Probe Interface unit</i>), NX02INT	44
2.6.23	Inlet temperature sensor (<i>optional</i>)	46
2.7	Final checks	47
3	Servo-motor selection and calibration	48
3.1	Servo-motor selection	48
3.1.1	Motor requirements	48
3.2	IMPORTANT: Servo-motor direction warning	49
3.3	Locking the servo-motor to the valve shaft	50
3.4	Feedback potentiometer	50
3.5	Adjusting microswitch positions	51
3.6	Servomotor Replacement	52
4	Description of operation	53
4.1	The Display / Keypad	53
4.2	Start-up sequence	54
4.3	Non-volatile lockout	59
4.4	Modulation	59
4.4.1	Normal/Remote mode.	59
4.4.2	Local mode. (LOC 1, LOC 2)	59
5	Commissioning the control	60
5.1	General	60
5.2	Commissioning data	61
5.2.1	Option parameters	61
5.2.2	Setpoints	61
5.3	The Commissioning process	61
5.3.1	Navigation	61
5.4	Description of Touchscreen operation.	63
5.4.1	The Touchscreen	63
5.4.2	Touchscreen Power on.	63
5.4.3	The Operation Mode line and Status messages	63
5.4.4	Touchscreen Overview display.	65

5.4.5	Touchscreen Lockout Reset (Alarm Mute) Function.	66
5.4.6	Touchscreen LED Indication.	66
5.4.7	Touchscreen Fuel/Profile Select.	67
5.4.8	Touchscreen 'Control' screen.	68
5.4.9	Touchscreen 'Menu' button.	68
5.4.10	Touchscreen 'Burner Settings' button.	68
5.4.11	Touchscreen 'Fault/Event Log' button.	70
5.4.12	Touchscreen 'Screen Configuration' button.	70
5.4.13	Cleaning the Touchscreen	71
5.4.14	Touchscreen Commission mode.	72
5.5	Using option set mode	74
5.5.1	Option parameter list	74
	Option 00.1 - Site passcode (0 - 999) DEFAULT VALUE is 154. LV3	74
	Option 00.2 - Serial communications control address (0 - 15) LV3	74
	Option 00.3 - Reset hours run (0 - 1) LV3	75
	Option 00.4 - User program select (0 / 1) LV1	75
	Option 00.6 - <i>Reserved</i>	75
	Option 00.7 - Modbus device address (1 - 99) LV3	75
	Option 00.8 - Modbus communications speed (0 - 3) LV3	76
	Option 01.0 - Power Up Option (0 - 30) LV3	76
	Option 01.1 - Keyboard Auto/Manual enable (0 - 1) LV3 [Engineer's Key 33]	76
	Option 01.2 - Mute/Reset Input enable (0 - 2) LV3	76
	 Option 02.x - Drive Name LV3	77
	 Option 03.x - Drive Serial Number LV3	78
	 Option 04.x - Drive Usage Information LV3	78
	 Option 05.x - Drive Options LV3	79
	 Options 06.1-06.4 - Profile Naming LV3	80
	Options 07.x, 08.x- <i>RESERVED</i>	80
	Option 09.0 - Inverter control accuracy (0 / 1) LV3	80
	Option 09.1 - Inverter error tolerance (0 / 1) LV3	80
	Option 09.2 - Inverter closed loop gain (15 - 125%) LV3	81
	Option 09.3 - Inverter stop time (0 - 100 seconds) LV3	81
	Option 09.4 - Inverter acceleration time (0-100 seconds) (60 seconds max. recommended) LV3	81
	Option 09.6 - VSD2 Speed Encoder Scaler (255 - 999) LV3	82
	Option 09.7 - <i>Reserved</i>	82
	Option 09.8 - <i>Reserved</i>	82
	Option 14.1 - Limit Relay (0 - 8) LV3	83
	Option 14.2 - Oxygen and Flue Temperature Limit Relay (0 - 8) LV3	83
	 Option 15.0 - Modulation sensor input type (0 - 3) LV3	84
	Option 15.1 - Modulation input decimal places (0 to 2)	84
	 Option 15.2 - Modulation input zero value (-999 to +999 / -99.9 to +99.9 / -9.99 to +9.99) LV3	85
	 Option 15.3 - Modulation input span value (-999 to +999 / -99.9 to +99.9 / -9.99 to +9.99) LV3	85
	 Option 15.4 - Setpoint display units (0 - 3) LV3	85
	 Option 15.5 - Boiler high safety limit (0 - 999 / 0.0 - 99.9 / 0.00 - 9.99) LV3	85
	Option 15.6 - Modulation Time (0 - 120 seconds) LV3	85
	Option 15.7 - Bumpless Transfer (0 or 1) LV3	86
	Option 15.8 - Low before Off (0 or 1) LV3	86
	Option 15.9 - <i>reserved</i>	86
	Option 16.2 - Allow profile swap (0 to 16) LV3	87
	Option 18.1 to 18.4 - Low Voltage Fail-safe Alarm / Lockout Inputs (0 - 460) LV3	91

Option 19.1 to 19.4 – Fail-safe Alarm / Lockout Display Messages LV3	93
Option 20.0 – Set-point Select input (0 - 35) LV3	94
Option 20.1 – Boiler Shutdown input (0 - 35). LV3	95
Option 20.2 – Low Fire Hold input (0 - 35). LV3	95
Option 20.3 – Oxygen Trim Disable input (0 - 35). LV3	95
Option 20.4 – Ignition Wait input (0 - 35). LV3	96
Option 20.5 – Purge Hold input (0 - 35). LV3	96
Option 20.6 – Purge Time Start input (0 - 35). LV3	96
Option 20.7 – Analog Input 5 function (0 - 2). LV3	96
Option 20.8 – Digital Input - Lead Boiler Select (0 - 35) LV3	97
 Option 21.0 – Set-point 1 enable (0 - 1) LV1	97
 Option 21.1 – Set-point 1 control value (0 - 999 / 00.0 - 99.9 / 0.00 - 9.99) LV1	97
 Option 21.2 – Set-point 1 proportional band (0 - 999 / 00.0 - 99.9 / 0.00 - 9.99) LV197	97
Option 21.3 – Set-point 1 integral time (0 - 999 seconds) LV1	97
Option 21.4 – Set-point 1 derivative time (0 - 999 seconds) LV1	98
 Option 21.5 - Setpoint 1 control limit type (0 - 2) LV3	98
Option 21.6 - Setpoint 1 low limit control value (Cut In) (0 - 999 / 00.0 - 99.9 / 0.00 - 9.99) LV1	98
Option 21.7 - Setpoint 1 high limit control value (Cut Out) (0 - 999 / 00.0 - 99.9 / 0.00 - 9.99) LV198	98
Option 21.8 – Remote Setpoint 1 zero (4ma) value (0 - 999 / 00.0 - 99.9 / 0.00 - 9.99) LV1	98
Option 21.9 – Remote Setpoint 1 span (20mA) value (0 - 999 / 00.0 - 99.9 / 0.00 - 9.99) LV1	98
Option 22.0 – PID (set-point) 2 enable (0 - 1) LV1	98
Option 22.1 – Set-point 2 control value (0 - 999 / 00.0 - 99.9 / 0.00 - 9.99) LV1	99
Option 22.2 – Set-point 2 proportional band (0 - 999 / 00.0 - 99.9 / 0.00 - 9.99) LV1	99
Option 22.3 – Set-point 2 integral term (0 - 999 seconds) LV1	99
Option 22.4 – Set-point 2 derivative term (0 - 100) LV1	99
Option 22.5 - Setpoint 2 control limit type (0 - 2) LV3	99
Option 22.6 - Setpoint 2 low limit control value (0 - 999 / 00.0 - 99.9 / 0.00 - 9.99) LV1	100
Option 22.7 - Setpoint 2 high limit control value (0 - 999 / 00.0 - 99.9 / 0.00 - 9.99) LV1	100
Option 22.8 – Maximum modulation rate	100
Option 22.9 – <i>Reserved</i>	100
Option 23.0 – Warming Enable (0 or 1) LV1	100
Option 23.1 – Warming Limit (0 - 999 / 00.0 - 99.9 / 0.00 - 9.99) LV1	100
Option 23.2 – Warming Time (0 to 999 minutes) LV1	100
Options 24.x – Sequencing	101
Option 24.0 Sequence Slaves (number of boilers) (0-3) LV1	102
Option 24.1 -24.3 Priority Number(s) LV1	102
Option 24.4 – Lead OFF Time LV1	103
Option 24.5 – Lead (master) Boiler Select Method LV1	103
Option 24.6 – lag (slave) On Rate (0-100%) LV1	104
Option 24.7 – lag (slave) On Delay (0-999 minutes) LV1	104
Option 24.8 – lag (slave) Off Rate (0-100%) LV3	104
Option 24.9 – lag (slave) Off Delay (0-999 minutes) LV3	104
Option 25.0 – Banking based on temperature / water temperature standby for steam boiler LV3104	104
Option 25.1 – Water temperature zero value LV3	105
Option 25.2 – Water temperature span value LV3	105
Option 25.3 – Water temperature cut-in LV1	105
Option 25.4 – Water temperature cut-out LV1	105
Option 25.5 – Temperature based warming limit LV1	105
Option 25.6 – Lag wait time LV1	105
Option 28.0 – Analog Input decimals (0-2) LV3	105
Option 28.1 – Analog Input 1 zero (0-999) LV3	105
Option 28.2 – Analog Input 1 span (0-999) LV3	105
Option 28.3 – Analog Input 2 decimals (0-2) LV3	105
Option 28.4 – Analog Input 2 zero (0-999) LV3	105
Option 28.5 – Analog Input 2 span (0-999) LV3	105



Option 28.6 – Analog Input 3 decimals (0-2) LV3	105
Option 28.7 – Analog Input 3 zero (0-999) LV3	106
Option 28.8 – Analog Input 3 span (0-999) LV3	106
Option 29.0 – 4-20mA Output 1 function if not VSD LV3	106
Option 29.1 – Output1 zero value (4mA output) LV3	106
Option 29.2 – Output1 span value (20mA output) LV3	106
Option 29.3 – 4-20mA Output 2 function if not VSD LV3	107
Option 29.4 – Output2 zero value (4mA output) LV3	107
Option 29.5 – Output2 span value (20mA output) LV3	107
Option 29.6 – 4-20mA Output 3 function if not VSD LV3	107
Option 29.7 – Output3 zero value (4mA output) LV3	107
Option 29.8 – Output3 span value (20mA output) LV3	107
Option 30.0 – Oxygen probe interface serial number LV3	109
Option 30.1 – Oxygen probe calibration offset value (0 – 999) LV3	109
Option 30.2 – Oxygen probe calibration gain value (0 – 999) LV3	110
Option 30.4 – Flue and inlet sensor temperature units (0 – 1) LV3	110
Option 30.5 – Oxygen input function (0/1) LV3	110
Option 30.6 – Oxygen probe calibrate enable (0 – 2) LV3	110
Option 30.7 – Boiler transport delay (5 – 60 seconds) LV3	111
Option 30.8 – Reset oxygen trim profile (0/1) LV3	111
Option 30.9 – Automatic trim commissioning (0 / 1) LV3	111
Options 31.1 to 31.4 – Trim Type for profiles 1 to 4 (0 to 2) LV3	113
Option 32.0 – Trim limit default (0/1) LV3	114
Options 33.1 to 33.4 – Trim integral gain (Default 10%) LV3	114
Option 34.0 – Trim proportional gain (0/1) LV3	115
Options 34.1 to 34.4 – Trim proportional gain (0.0 – 99.9%) LV3	115
Option 34.5 – Calculating and entering the flow values manually LV3	115
Option 35.0 – Inlet temperature sensor serial number LV3	118
Options 35.5 to 35.8 – Calorific values of fuels/profiles 1 to 4 respectively (0.0 – 99.9) LV3	118
Option 35.9 – Boiler radiated heat loss (0.0 – 9.9%) LV3	119
Option 36.0 – Flue temperature alarm select (0/1) LV3	119
Options 36.1 to 36.4 – Flue temperature low alarm values (0 – 999) LV3	119
Option 38.0 – Oxygen alarm select (0 – 2) LV3	120
Options 38.1 to 38.4 – Oxygen low alarm values at low fire (0.0 – 99.9%) LV3	120
Options 39.1 to 39.4 – Oxygen low alarm values at high fire (0.0 – 99.9%) LV3	120
Options 40.1 to 40.4 – Oxygen high alarm values at low fire (0.0 – 99.9%) LV3	120
Options 41.1 to 41.4 – Oxygen high alarm values at high fire (0.0 – 99.9%) LV3	120
Option 42.0 – Second Oxygen probe interface serial number LV3	121
Option 42.1 – Second Oxygen probe calibration offset value (0 – 999) LV3	121
Option 42.2 – Second Oxygen probe calibration gain value (0 – 999) LV3	121
Option 42.3 – Second Oxygen probe calibrate enable (0 – 2) LV3	122
Option 42.4 – Max oxygen variation (0 – 9.9 %) LV3	122
Option 42.5 – Max flue temp variation (0 to 999 °C) LV3	123
Option 42.7 – Air pressure monitoring* – input number LV3	123
Option 42.8 – Air pressure monitoring* – Span LV3	123
Option 42.9 – Air pressure monitoring* – max error LV3	123
Option 43.0 – Water level / expansion unit** LV3 - <i>Reserved</i>	123
Option 43.1 – Water level / expansion unit** LV3 - <i>Reserved</i>	123
Option 44.0 – Set real time clock (0/1) LV1	123
Option 44.1 – Set Year (00 – 99) LV1	123
Option 44.2 – Set Month (1 – 12) LV1	123
Option 44.3 – Set Day of Month (1 – 31) LV1	123
Option 44.4 – Set Day of Week (1 - 7) LV1	124
Option 44.5 – Set Hours (0 - 23) LV1	124
Option 44.6 – Set Minutes (0 - 59) LV1	124
Option 44.7 – Set Seconds (0 - 59) LV1	124



Option 44.9 – Reset Fault Log (0 - 1) LV3	124
Option 45.0 - Erase / Restore enable (0 - 1) LV3	124
Option 45.1 - Erase command (0 - 5) LV3	125
Option 45.2 - Restore command (0 - 999) LV3	125
5.6 Commission ratio mode	126
5.6.1 Description	126
5.6.2 Using commission ratio mode	126
5.6.3 Leaving commission ratio mode	128
5.7 Adjust ratio mode	128
5.7.1 Description	128
5.7.2 Entering adjust ratio mode	128
5.7.3 Using adjust ratio mode	129
5.7.4 Leaving adjust ratio mode	130
5.8 Interrogating the software issue	130
6 Faults and fault finding	131
6.1 The fault display	131
6.2 What to do when a fault occurs	131
6.3 Non-volatile lockout	132
6.4 Fault subsets	132
6.5 Fault listing	132
6.6 The engineer's key	139
6.6.1 Fault history	140
6.6.2 Engineer's key parameter list	140
6.7 Troubleshooting	150
6.7.1 Display / General	150
6.7.2 Startup	151
6.7.3 Commissioning	151
6.7.4 Modulation	152
6.7.5 Inverters	153
6.8 YB110/TSD104 Fault Messages	155
7 Inverter (variable speed) drives	157
7.1 System configuration	157
7.2 Description of control method	158
7.3 Setting up the inverter for use with the PPC6000 series	158
7.4 Setting up the PPC6000 series for use with a inverter drive	159
8 The oxygen trim option	160
8.1 Oxygen Pro probe description	160
8.1.1 Installation of oxygen probe	160
8.1.2 Mounting the oxygen probe	160
9 Appendix	163
9.1 Calibrating and servicing the oxygen probe	163
9.1.1 Probe calibration	163
9.1.2 Oxygen Probe Filter Testing	164
9.1.3 Removing the oxygen probe from the flue	164
9.1.4 Filter replacement	165
9.1.5 Probe mounted flue thermocouple replacement.	165
9.2 Troubleshooting inverter problems	167
9.3 PID Tutorial	168



9.4	Combustion Profile Setup Guideline	169
9.5	Fireye PPC6000 Efficiency Calculations	170
9.6	Typical Wiring Diagrams	172
9.7	Operator's Guide	179
9.7.1	Introduction	179
9.7.2	General	179
9.7.3	Password	179
9.7.4	Adjusting Cut IN and Cut OUT	180
9.7.5	Manual Modulation	181
9.7.6	Resetting the Control	181
9.7.7	Fault and Fault Finding	182
9.7.8	Fault subsets	182
9.7.9	Fault listing	183
9.7.10	PID Tutorial	183
9.7.11	The startup / shutdown stages are as follows:	184
9.7.12	Non-volatile lockout	187
9.8	Fireye Nexus Combustion Curve	188

10	Function Block Programming Details	189
	Introduction to Fireye's Abacus Programming Software	189
1.1.	Introduction	189
1.4.	The schematic toolbar	193
1.5.	Place Function blocks	194
1.6.	Moving and editing the link lines	194

11	Circuit board connections	196
11.1	Inlet Air Temperature Sensor (Optional)	196
11.2	PPC6000 Variable Speed Drive Daughter Board P/N NXDBVSD	197
11.3	Oxygen probe connection (optional Oxygen Probe Interface unit)	198
11.4	PPC6000	199